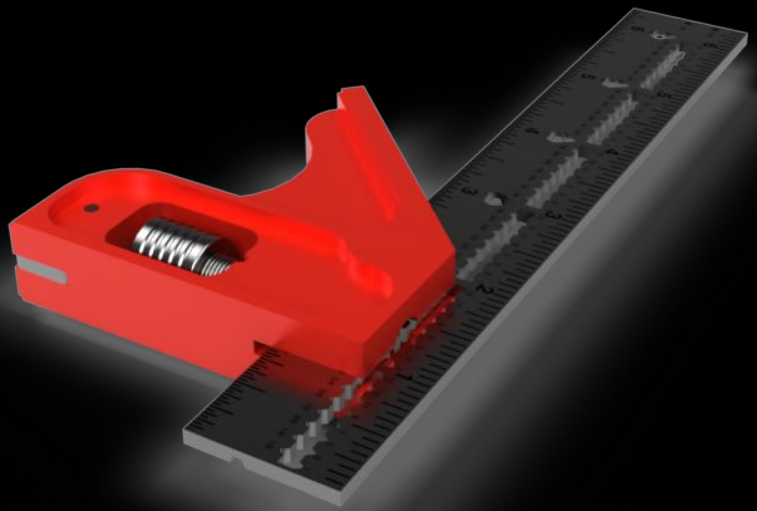





# CAD Portfolio

Neil G. Senecal

*nsenecal@byu.edu*

*nsenecal.github.io*

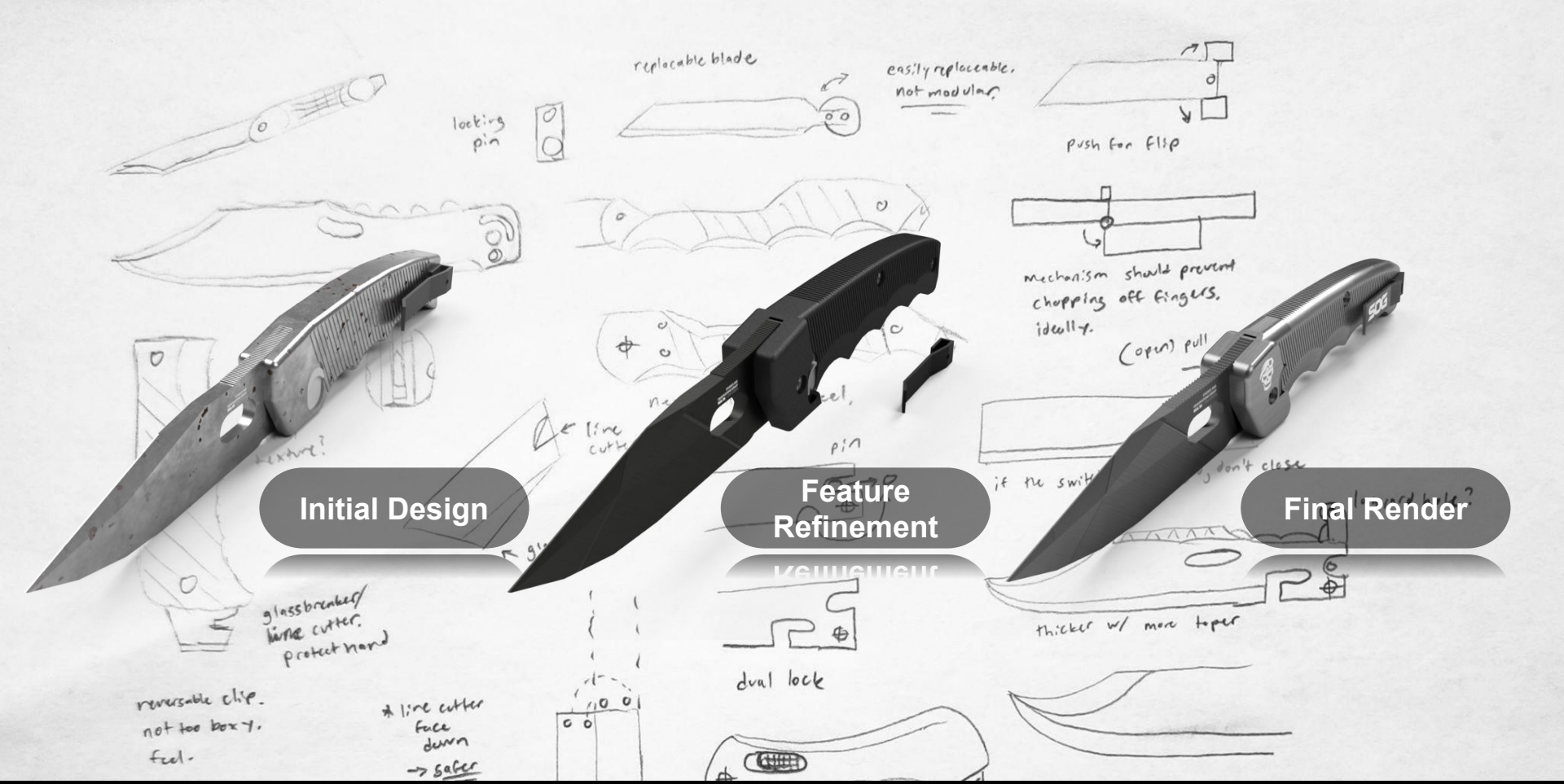
# SOG Survival Knife

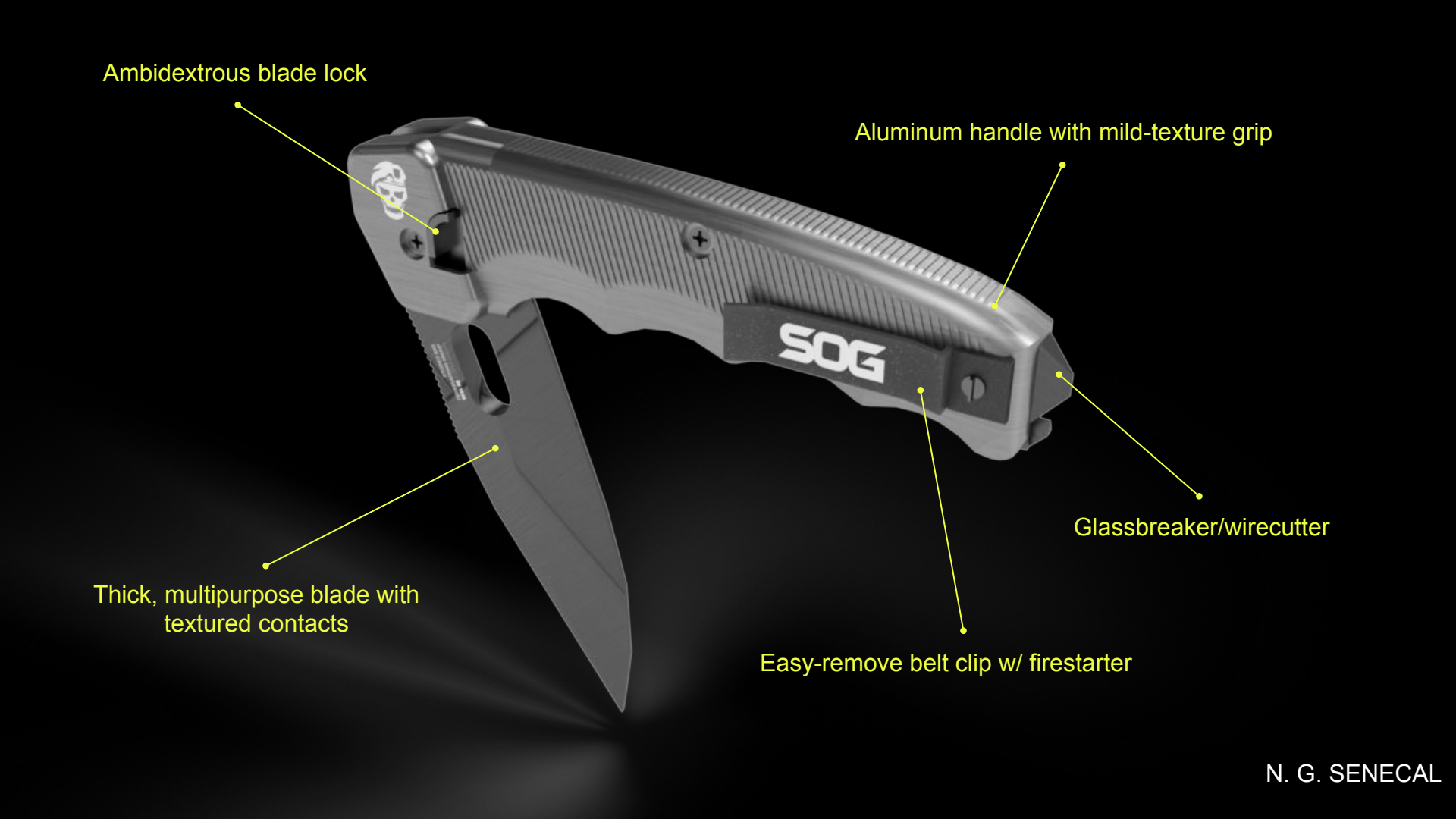
*NEXT GEN DESIGN CHALLENGE*
*26 Hours*

Participants were directed to select a product and redesign it with the intent of making it “next gen.”




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replaceable blade
easily replaceable, not modular
push for flip
locking pin
mechanism should prevent chopping off fingers, ideally.
(open) pull
pin
thicker w/ more taper
dual lock
glassbreaker/ little cutter. protect hand
reversable clip. not too boxy. feel.
* line cutter face down → safer
Initial Design
Feature Refinement
Final Render


Ambidextrous blade lock
Aluminum handle with mild-texture grip
SOG
Glassbreaker/wirecutter
Thick, multipurpose blade with textured contacts
Easy-remove belt clip w/ firestarter

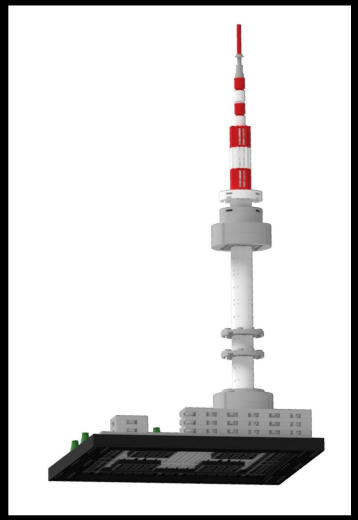
N. G. SENECAL

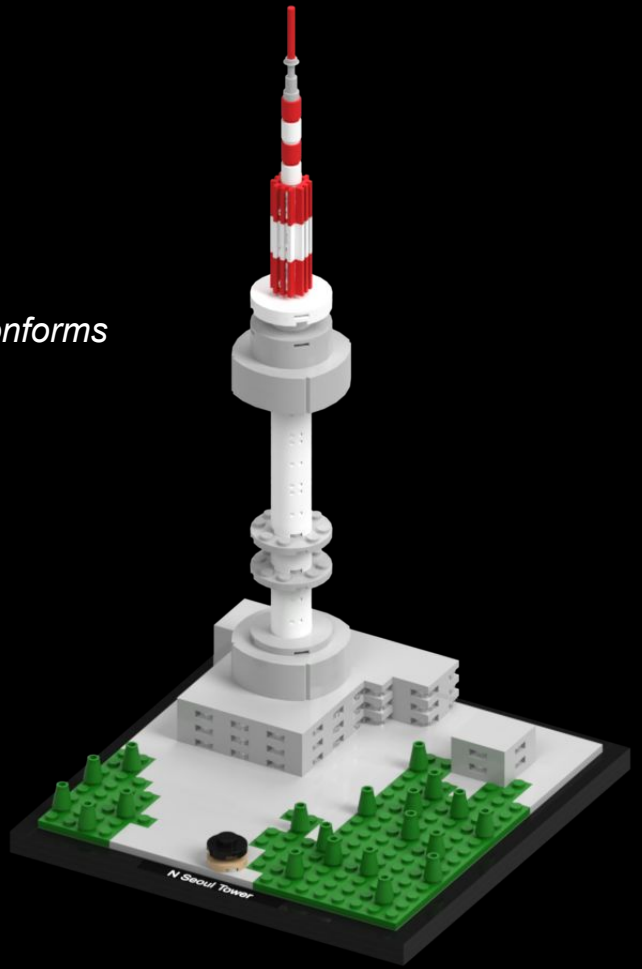
# North Seoul Tower (Namsan Tower)

*LEGO DESIGN CHALLENGE*
*27 Hours*

*Participants were tasked with designing a LEGO structure that conforms to the LEGO Group's standards aesthetics and set design.*

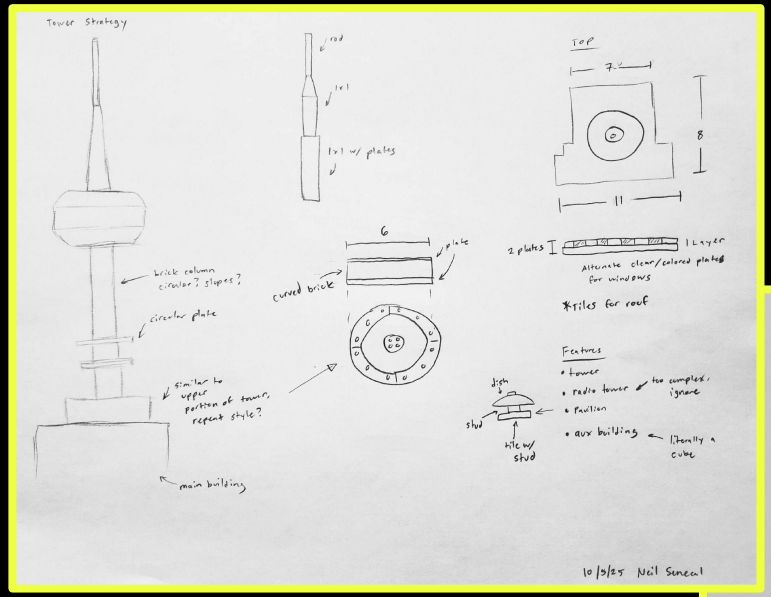


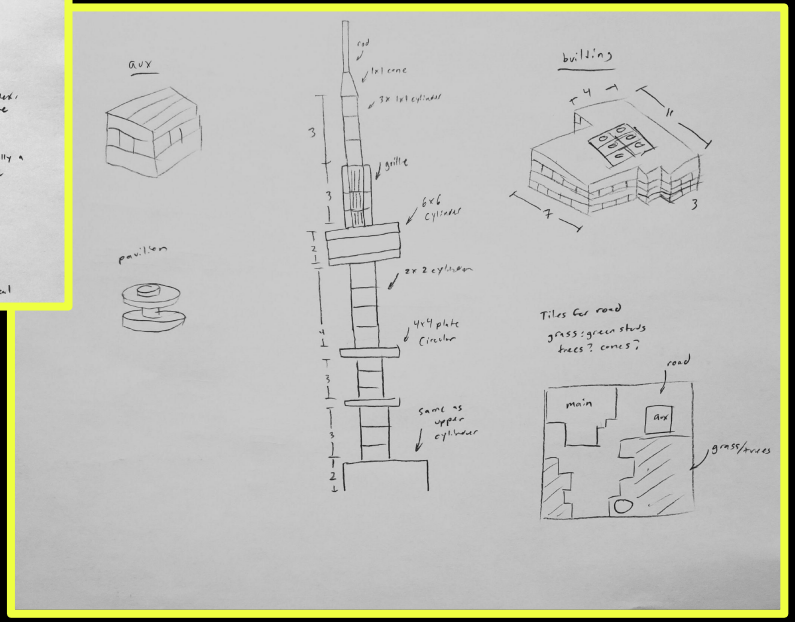




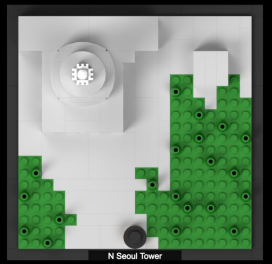

N. G. SENECAL

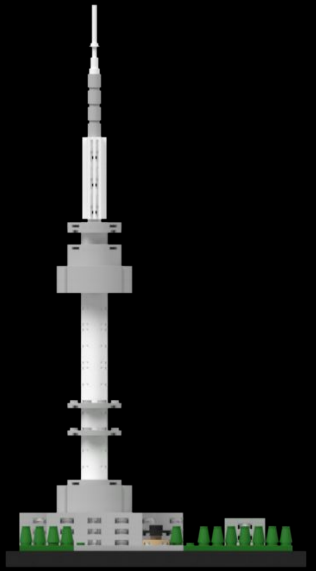
## Initial Sketches/Strategies




## Preliminary Renders



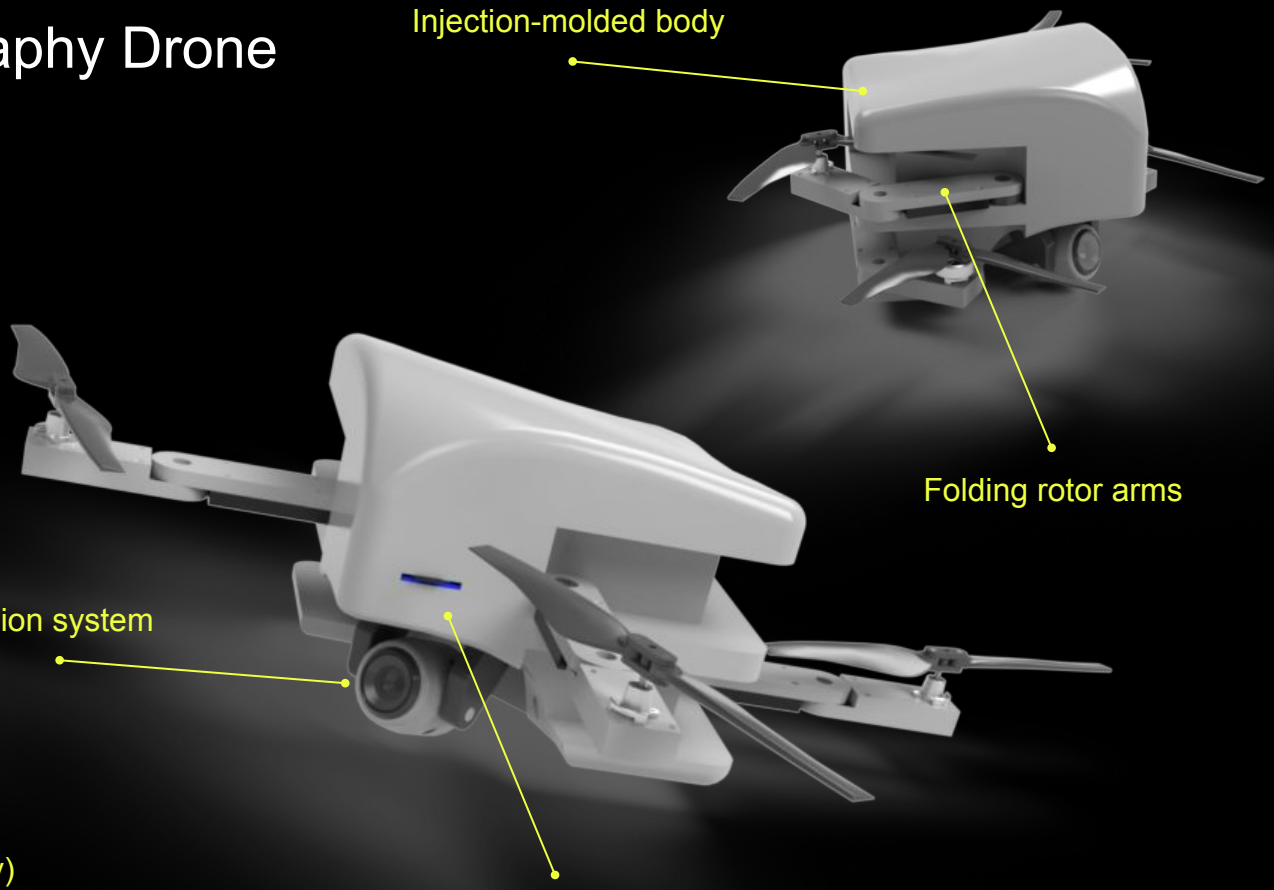
N. G. SENECAL

# Compact Videography Drone

Vision and Optics Module

*TEAM DESIGN CHALLENGE*
*22 Hours (Individual)*

Based on the needs of a professional videographer, teams were tasked with designing a compact quadcopter for capturing in-field footage.



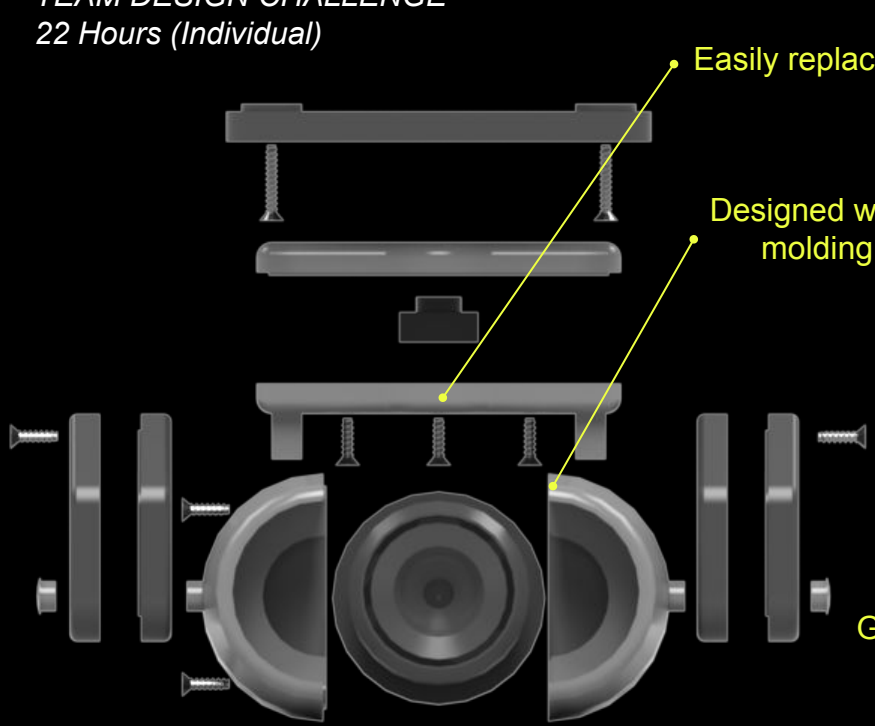

Designed in collaboration with:
Holland Tanner (arm assembly)
Conner Rayburn (body assembly)

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# Compact Videography Drone

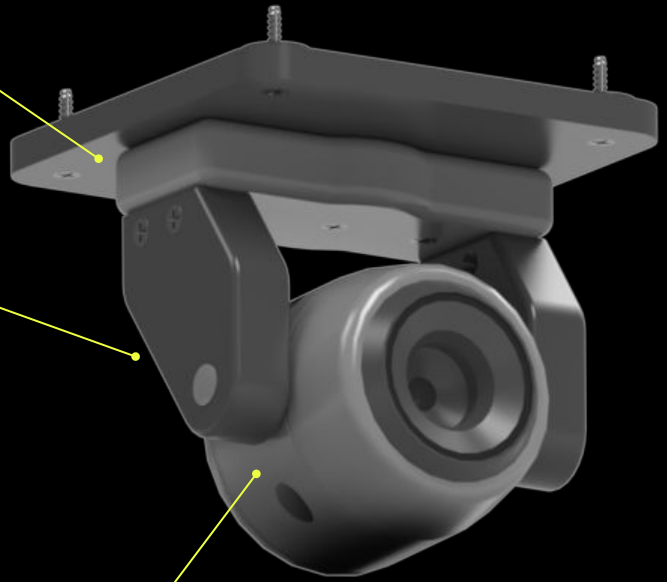
Vision and Optics Module

*TEAM DESIGN CHALLENGE*
*22 Hours (Individual)*



Easily replaceable modules

Designed with injection molding in mind



Gimbaled mount with 6 DOF

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